Автономная некоммерческая образовательная организация высшего образования Центросоюза Российской Федерации “Сибирский университет потребительской кооперации “ (СибУПК)

Отчёт

По лабораторной работе №3

“Работа с базовой системой ввода-вывода (BIOS) и диагностика накопителей (SMART)”

Дисциплина: Инженерно-техническая поддержка ИС

Выполнил студент: Клименков Егор

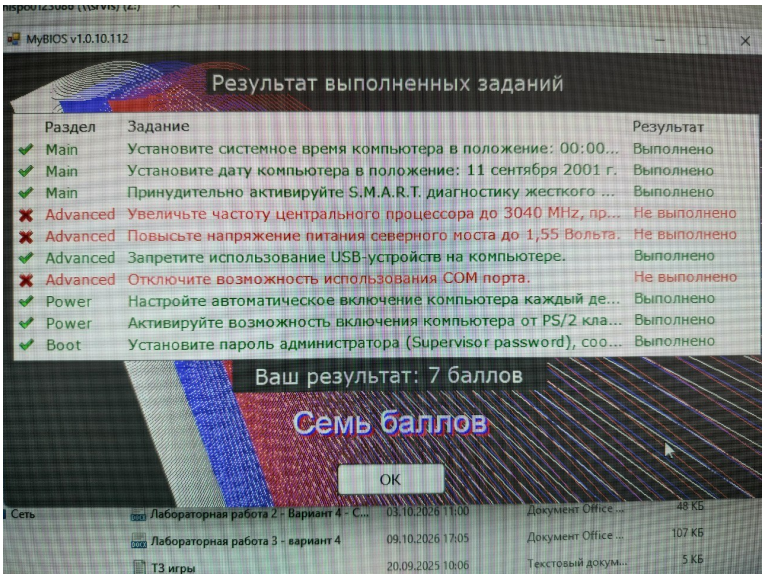
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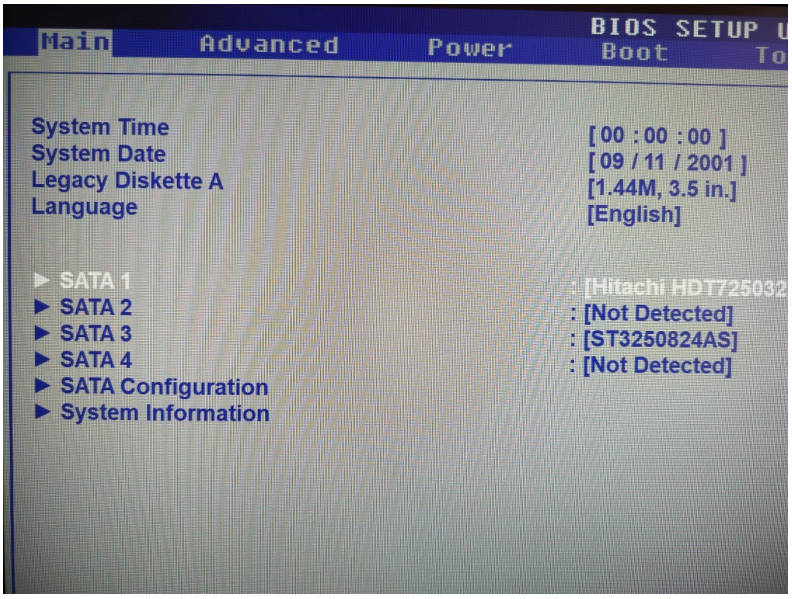
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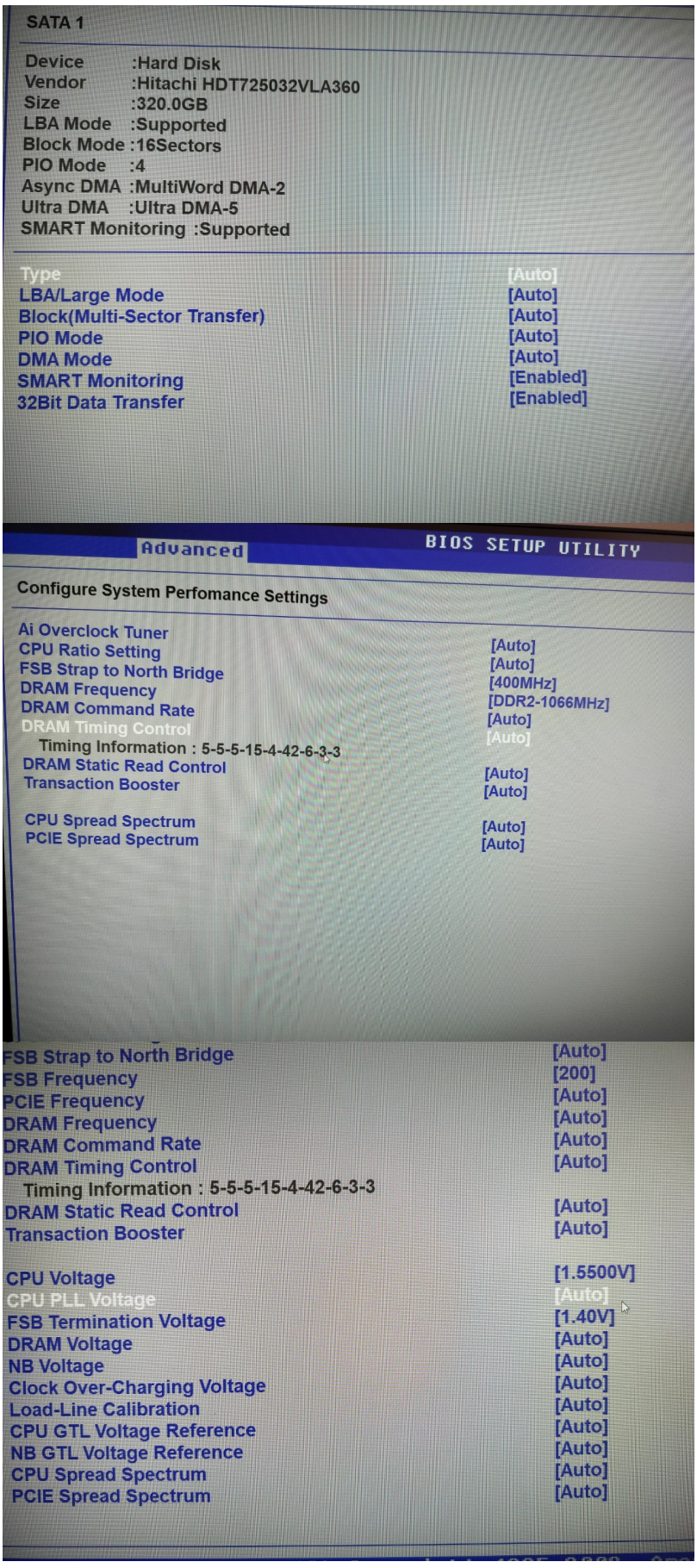
Шифр студента: ispo-0123086

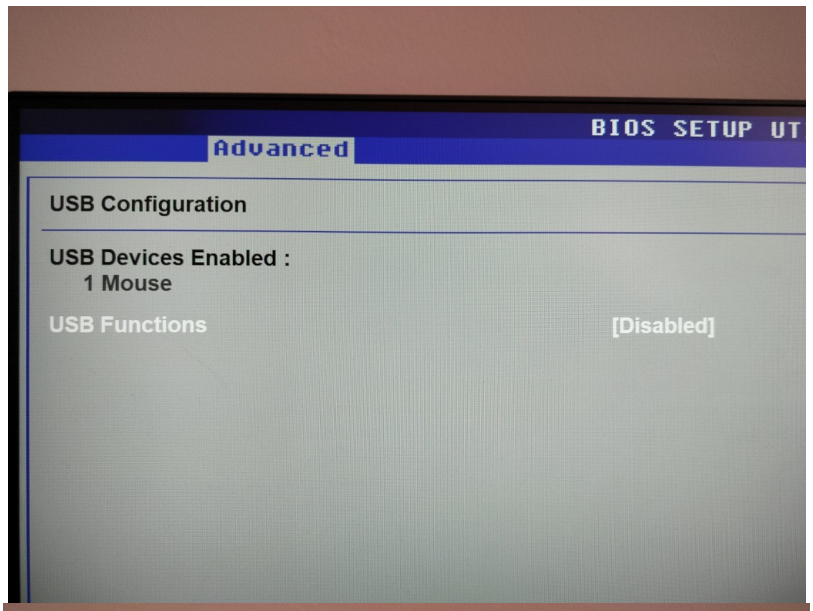
Г. Новосибирск, 2026 г.

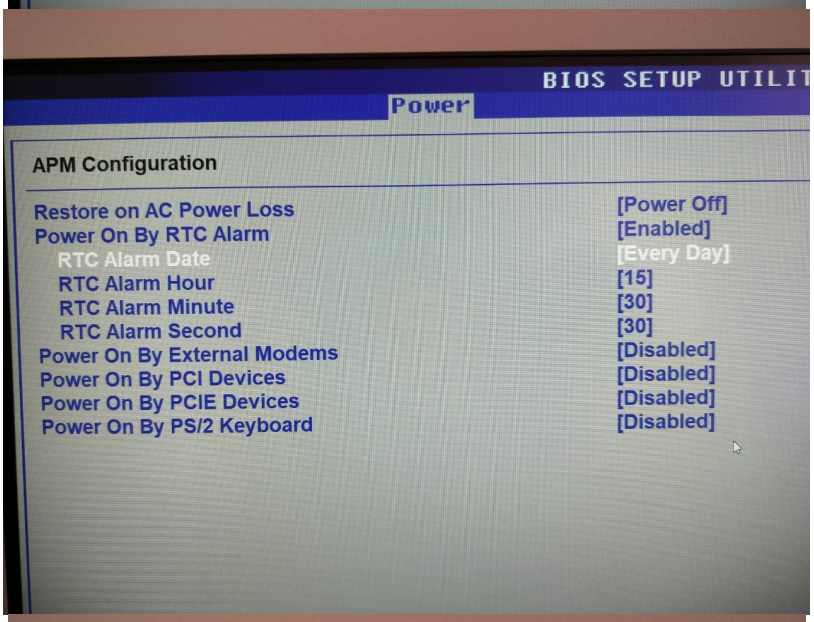
Раздел 1. Тренажёр BIOS MyBIOS

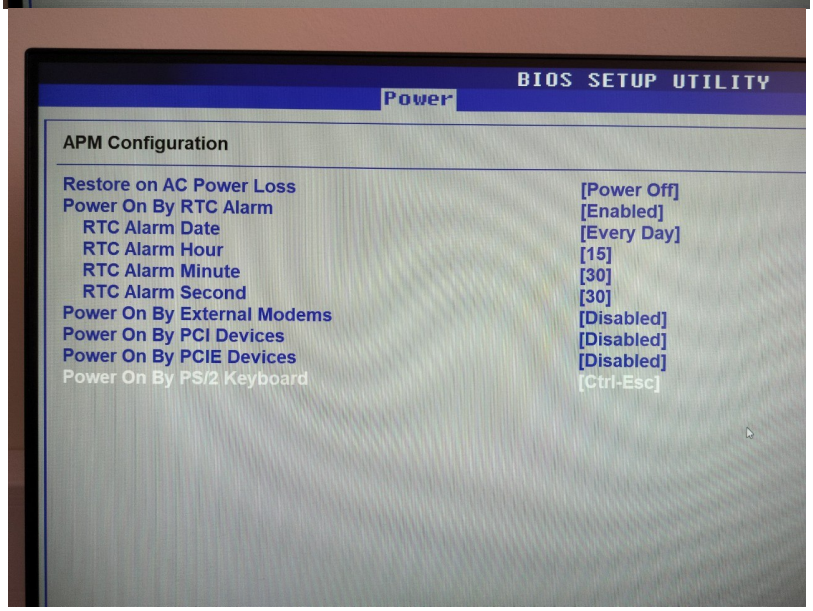


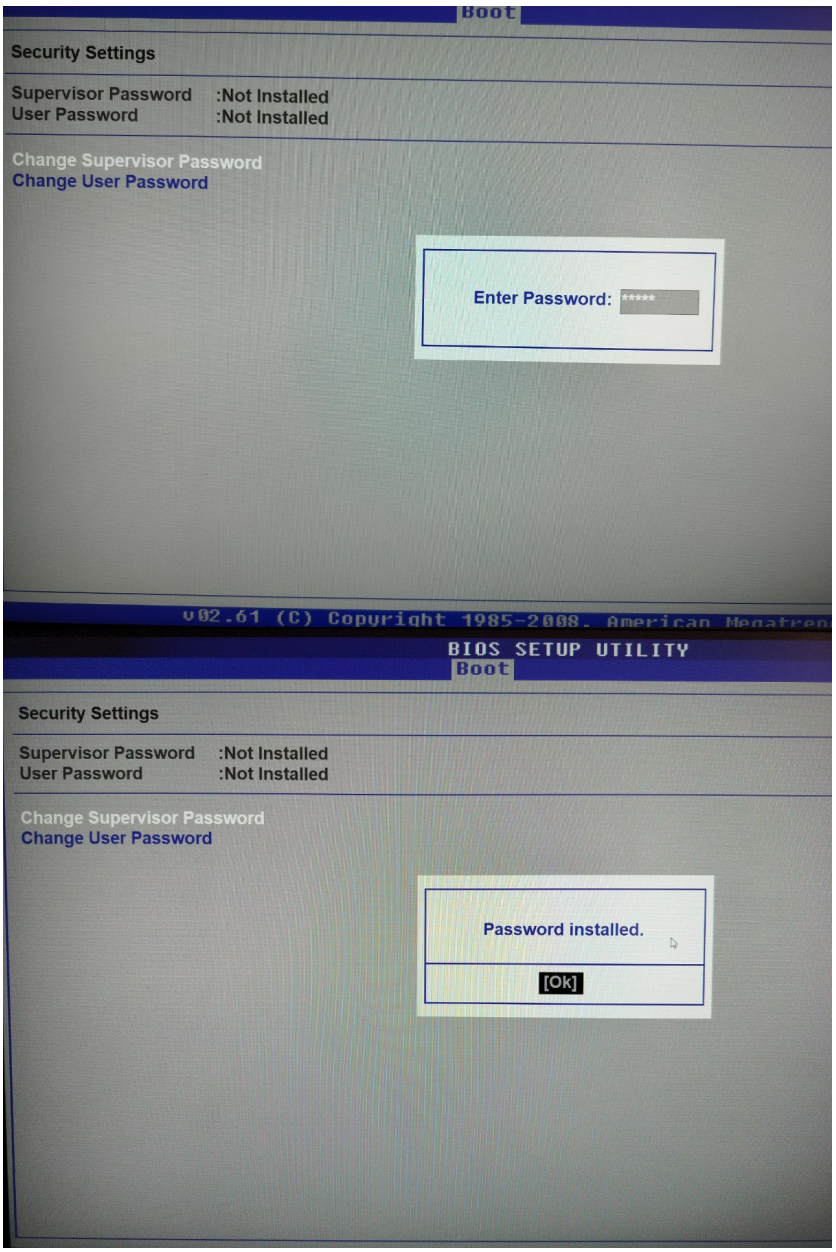



SATA 1
Device :Hard Disk
Vendor :Hitachi HDT725032VLA360
Size :320.0GB
LBA Mode :Supported
Block Mode :16Sectors
PIO Mode :4
Async DMA :MultiWord DMA-2
Ultra DMA :Ultra DMA-5
SMART Monitoring :Supported
Type [Auto]
LBA/Large Mode [Auto]
Block(Multi-Sector Transfer) [Auto]
PIO Mode [Auto]
DMA Mode [Auto]
SMART Monitoring [Enabled]
32Bit Data Transfer [Enabled]
Advanced
BIOS SETUP UTILITY
Configure System Perfomance Settings
Ai Overclock Tuner [Auto]
CPU Ratio Setting [Auto]
FSB Strap to North Bridge [400MHz]
DRAM Frequency [DDR2-1066MHz]
DRAM Command Rate [Auto]
DRAM Timing Control [Auto]
Timing Information : 5-5-5-15-4-42-6-3-3
DRAM Static Read Control [Auto]
Transaction Booster [Auto]
CPU Spread Spectrum [Auto]
PCIE Spread Spectrum [Auto]
FSB Strap to North Bridge [Auto]
FSB Frequency [200]
PCIE Frequency [Auto]
DRAM Frequency [Auto]
DRAM Command Rate [Auto]
DRAM Timing Control [Auto]
Timing Information : 5-5-5-15-4-42-6-3-3
DRAM Static Read Control [Auto]
Transaction Booster [Auto]
CPU Voltage [1.5500V]
CPU PLL Voltage [Auto]
FSB Termination Voltage [1.40V]
DRAM Voltage [Auto]
NB Voltage [Auto]
Clock Over-Charging Voltage [Auto]
Load-Line Calibration [Auto]
CPU GTL Voltage Reference [Auto]
NB GTL Voltage Reference [Auto]
CPU Spread Spectrum [Auto]
PCIE Spread Spectrum [Auto]


BIOS SETUP UT
Advanced
USB Configuration
USB Devices Enabled :
1 Mouse
USB Functions
[Disabled]


BIOS SETUP UTILIT
Power
APM Configuration
Restore on AC Power Loss
[Power Off]
Power On By RTC Alarm
[Enabled]
RTC Alarm Date
[Every Day]
RTC Alarm Hour
[15]
RTC Alarm Minute
[30]
RTC Alarm Second
[30]
Power On By External Modems
[Disabled]
Power On By PCI Devices
[Disabled]
Power On By PCIE Devices
[Disabled]
Power On By PS/2 Keyboard
[Disabled]


BIOS SETUP UTILITY
Power
APM Configuration
Restore on AC Power Loss
[Power Off]
Power On By RTC Alarm
[Enabled]
RTC Alarm Date
[Every Day]
RTC Alarm Hour
[15]
RTC Alarm Minute
[30]
RTC Alarm Second
[30]
Power On By External Modems
[Disabled]
Power On By PCI Devices
[Disabled]
Power On By PCIE Devices
[Disabled]
Power On By PS/2 Keyboard
[Ctrl-Esc]


Boot
Security Settings
Supervisor Password :Not Installed
User Password :Not Installed
Change Supervisor Password
Change User Password
Enter Password: *****
v02.61 (C) Copyright 1985-2008, American Megatren
BIOS SETUP UTILITY
Boot
Security Settings
Supervisor Password :Not Installed
User Password :Not Installed
Change Supervisor Password
Change User Password
Password installed.
[Ok]

Раздел 2. Диагностика накопителей

Использованная утилита:

консольная утилита `smartctl` (из пакета `smartmontools`) в операционной системе Linux (CachyOS).

Модели накопителя для диагностики:

KINGSTON SFYRS1000G (NVMe M.2 SSD)

WDC WD10EZEX-07WN4A0 (HDD Western Digital Blue, 1 ТБ)

NVMe M.2**:**

```
❯ sudo smartctl -a /dev/nvme0n1
smartctl 7.5 2025-04-30 r5714 [x86_64-linux-7.2.11-1-cachyos] (local build)
Copyright (C) 2002-25, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Model Number:                       KINGSTON SFYRS1000G
Serial Number:                      50026B7382CC6DB0
Firmware Version:                   EIFK31.7
PCI Vendor/Subsystem ID:            0x2646
IEEE OUI Identifier:                0x0026b7
Total NVM Capacity:                 1 000 204 886 016 [1,00 TB]
Unallocated NVM Capacity:           0
Controller ID:                      1
NVMe Version:                       1.4
Number of Namespaces:               1
Namespace 1 Size/Capacity:          1 000 204 886 016 [1,00 TB]
Namespace 1 Formatted LBA Size:     512
Namespace 1 IEEE EUI-64:            0026b7 382cc6db05
Local Time is:                      Thu Oct 22 17:49:31 2026 +07
Firmware Updates (0x12):            1 Slot, no Reset required
Optional Admin Commands (0x0017):   Security Format Frmw_DL Self_Test
Optional NVM Commands (0x005d):     Comp DS_Mngmt Wr_Zero Sav/Sel_Feat Timestmp
Log Page Attributes (0x08):         Telmtry_Lg
Maximum Data Transfer Size:         512 Pages
Warning  Comp. Temp. Threshold:     84 Celsius
Critical Comp. Temp. Threshold:     89 Celsius

Supported Power States
St Op     Max   Active     Idle   RL RT WL WT  Ent_Lat  Ex_Lat
 0 +     8.80W       -        -    0  0  0  0        0       0
 1 +     7.10W       -        -    1  1  1  1        0       0
 2 +     5.20W       -        -    2  2  2  2        0       0
 3 -   0.0620W       -        -    3  3  3  3     2500    7500
 4 -   0.0620W       -        -    4  4  4  4     2500    7500

Supported LBA Sizes (NSID 0x1)
Id Fmt  Data  Metadt  Rel_Perf
 0 +     512       0         2
 1 -    4096       0         1

=== START OF SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

SMART/Health Information (NVMe Log 0x02, NSID 0xffffffff)
Critical Warning:                   0x00
Temperature:                        34 Celsius
Available Spare:                    100%
Available Spare Threshold:          10%
Percentage Used:                    4%
Data Units Read:                    206 760 872 [105 TB]
```

```
Data Units Written:                 97 597 421 [49,9 TB]
Host Read Commands:                 830 313 538
Host Write Commands:                315 553 635
Controller Busy Time:               1 639
Power Cycles:                       943
Power On Hours:                     8 392
Unsafe Shutdowns:                   71
Media and Data Integrity Errors:    0
Error Information Log Entries:      3 824
Warning  Comp. Temperature Time:    0
Critical Comp. Temperature Time:    0
Temperature Sensor 2:               66 Celsius

Error Information (NVMe Log 0x01, 16 of 63 entries)
Num   ErrCount  SQId   CmdId  Status  PELoc          LBA  NSID    VS  Message
  0       3824     0  0x1008  0x4004  0x028            0     1     -  Invalid Field in Command
  1       3823     0  0x000c  0x4004  0x028            0     0     -  Invalid Field in Command
  2       3822     0  0x0012  0x4004      -            0     0     -  Invalid Field in Command

Self-test Log (NVMe Log 0x06, NSID 0xffffffff)
Self-test status: No self-test in progress
No Self-tests Logged
```

WD10EZEX:

```
❯ sudo smartctl -a /dev/sda
smartctl 7.5 2025-04-30 r5714 [x86_64-linux-7.2.11-1-cachyos] (local build)
Copyright (C) 2002-25, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Model Family:     Western Digital Blue (CMR)
Device Model:     WDC WD10EZEX-07WN4A0
Serial Number:    WD-WCC6Y1LSA133
LU WWN Device Id: 5 0014ee 212c97599
Firmware Version: 01.01A01
User Capacity:    1 000 204 886 016 bytes [1,00 TB]
Sector Sizes:     512 bytes logical, 4096 bytes physical
Rotation Rate:    7200 rpm
Form Factor:      3.5 inches
Device is:        In smartctl database 7.5/5706
ATA Version is:   ACS-3 T13/2161-D revision 3b
SATA Version is:  SATA 3.1, 6.0 Gb/s (current: 6.0 Gb/s)
Local Time is:    Thu Oct 22 17:51:35 2026 +07
SMART support is: Available - device has SMART capability.
SMART support is: Enabled

=== START OF READ SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

General SMART Values:
Offline data collection status:  (0x82) Offline data collection activity
                                        was completed without error.
                                        Auto Offline Data Collection: Enabled.
Self-test execution status:      (   0) The previous self-test routine completed
                                        without error or no self-test has ever
                                        been run.
Total time to complete Offline
data collection:                (10920) seconds.
Offline data collection
capabilities:                    (0x7b) SMART execute Offline immediate.
                                        Auto Offline data collection on/off support.
                                        Suspend Offline collection upon new
                                        command.
                                        Offline surface scan supported.
                                        Self-test supported.
                                        Conveyance Self-test supported.
                                        Selective Self-test supported.
SMART capabilities:            (0x0003) Saves SMART data before entering
                                        power-saving mode.
                                        Supports SMART auto save timer.
Error logging capability:        (0x01) Error logging supported.
                                        General Purpose Logging supported.
Short self-test routine
recommended polling time:        (   2) minutes.
```

```
Extended self-test routine
recommended polling time:        ( 114) minutes.
Conveyance self-test routine
recommended polling time:        (   5) minutes.
SCT capabilities:              (0x3035) SCT Status supported.
                                        SCT Feature Control supported.
                                        SCT Data Table supported.

SMART Attributes Data Structure revision number: 16
Vendor Specific SMART Attributes with Thresholds:
ID# ATTRIBUTE_NAME          FLAG     VALUE WORST THRESH TYPE      UPDATED  WHEN_FAILED RAW_VALUE
  1 Raw_Read_Error_Rate     0x002f   200   200   051    Pre-fail  Always       -       0
  3 Spin_Up_Time            0x0027   174   171   021    Pre-fail  Always       -       2300
  4 Start_Stop_Count        0x0032   095   095   000    Old_age   Always       -       5656
  5 Reallocated_Sector_Ct   0x0033   200   200   140    Pre-fail  Always       -       0
  7 Seek_Error_Rate         0x002e   200   200   000    Old_age   Always       -       0
  9 Power_On_Hours          0x0032   068   068   000    Old_age   Always       -       24015
 10 Spin_Retry_Count        0x0032   100   100   000    Old_age   Always       -       0
 11 Calibration_Retry_Count 0x0032   100   100   000    Old_age   Always       -       0
 12 Power_Cycle_Count       0x0032   098   098   000    Old_age   Always       -       2877
192 Power-Off_Retract_Count 0x0032   200   200   000    Old_age   Always       -       159
193 Load_Cycle_Count        0x0032   193   193   000    Old_age   Always       -       21823
194 Temperature_Celsius     0x0022   110   090   000    Old_age   Always       -       33
196 Reallocated_Event_Count 0x0032   200   200   000    Old_age   Always       -       0
197 Current_Pending_Sector  0x0032   200   200   000    Old_age   Always       -       0
198 Offline_Uncorrectable   0x0030   200   200   000    Old_age   Offline      -       0
199 UDMA_CRC_Error_Count    0x0032   200   200   000    Old_age   Always       -       1
200 Multi_Zone_Error_Rate   0x0008   200   200   000    Old_age   Offline      -       0

SMART Error Log Version: 1
No Errors Logged

SMART Self-test log structure revision number 1
No self-tests have been logged.  [To run self-tests, use: smartctl -t]

SMART Selective self-test log data structure revision number 1
 SPAN  MIN_LBA  MAX_LBA  CURRENT_TEST_STATUS
    1        0        0  Not_testing
    2        0        0  Not_testing
    3        0        0  Not_testing
    4        0        0  Not_testing
    5        0        0  Not_testing
Selective self-test flags (0x0):
  After scanning selected spans, do NOT read-scan remainder of disk.
If Selective self-test is pending on power-up, resume after 0 minute delay.

The above only provides legacy SMART information - try 'smartctl -x' for more
```

Аналитический вывод о состоянии и ресурсе накопителя:

Накопитель №1 (SSD Kingston NVMe):

- **Состояние:** Исправен, критических ошибок нет
- **Температура:** 34 °C
- **Ресурс:** Наработка составляет 8 392 часа. Износ ячеек памяти - 4%, что свидетельствует об огромном остатке ресурса. Ошибок целостности данных нет.

Накопитель №2 (HDD Western Digital Blue):

- **Состояние:** Исправен
- **Температура:** 33 °C
- **Ресурс и надежность:** Время наработки составляет 24 015 часов. отсутствие поврежденных секторов. Механика работает стабильно